

Retail Sales & Profit Intelligence Dashboard

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Batch No : FNB16

Course : AI-driven Data Analytics

Project Objective

- Clean and organize raw data.
- Handle missing and inconsistent values.
- Integrate multiple tables using XLOOKUP/VLOOKUP.
- Calculate key business metrics such as Sales, Cost, Profit, and Discount Value.
- Perform descriptive statistical analysis.
- Create Pivot Tables for business analysis.
- Develop an interactive dashboard using charts and slicers.
- Generate insights to support business decision-making.

Project Description

This project focuses on cleaning, integrating, analyzing, and visualizing an e-commerce retail dataset using Microsoft Excel. The objective is to transform raw transactional data into meaningful business insights through data cleaning, Excel formulas, Pivot Tables, descriptive statistics, and an interactive dashboard. The project helps management understand sales performance, profitability, customer behavior, and regional trends.

Dataset Overview

The project uses the following tables:

- Sales_Fact
- Customer_Dim
- Product_Dim
- Store_Dim
- Integrated Dataset

(The tables were combined to create a complete dataset for analysis and reporting.)

Quantity	Unit_Price	Discount	Discount value	Payment_Type	Total_Amount	Cost	Total Cost	Total Profit
1	\$218.04	0.09	\$19.62	Credit Card	\$198.42	\$245.98	\$245.98	-\$47.56
3	\$104.47	0.27	\$28.21	PayPal	\$228.79	\$403.71	\$1,211.13	-\$982.34
1	\$278.63	0.07	\$19.50	Credit Card	\$259.13	\$166.06	\$166.06	\$93.07
2	\$401.50	0.18	\$72.27	Credit Card	\$658.46	\$179.77	\$359.54	\$298.92
2	\$909.51	0.27	\$245.57	Credit Card	\$1,327.88	\$17.31	\$34.62	\$1,293.26
4	\$593.84	0.24	\$142.52	PayPal	\$1,805.27	\$218.22	\$872.88	\$932.39
3	\$909.11	0.23	\$209.10	COD	\$2,100.04	\$107.06	\$321.18	\$1,778.86
4	\$996.85	0.24	\$239.24	PayPal	\$3,030.42	\$166.06	\$664.24	\$2,366.18
1	\$522.69	0.06	\$31.36	COD	\$491.33	\$112.42	\$112.42	\$378.91
3	\$261.05	0.12	\$31.33	PayPal	\$689.17	\$434.84	\$1,304.52	-\$615.35
3	\$644.73	0.01	\$6.45	COD	\$1,914.85	\$255.57	\$766.70	\$1,148.15
2	\$541.52	0.26	\$140.80	PayPal	\$801.45	\$158.14	\$316.28	\$485.17
1	\$395.18	0.09	\$35.57	PayPal	\$359.61	\$266.86	\$266.86	\$92.75
4	\$56.37	0.17	\$9.58	Credit Card	\$187.15	\$439.07	\$1,756.28	-\$1,569.13
3	\$359.56	0.17	\$61.13	PayPal	\$895.30	\$174.02	\$522.06	\$373.24
4	\$202.59	0.06	\$12.16	COD	\$761.74	\$228.53	\$914.12	-\$152.38
2	\$358.36	0.05	\$17.92	PayPal	\$680.88	\$285.54	\$571.08	\$109.80
4	\$85.62	0.28	\$23.97	COD	\$246.59	\$353.47	\$1,413.88	-\$1,167.29
4	\$929.26	0.23	\$213.73	PayPal	\$2,862.12	\$218.86	\$875.44	\$1,986.68
2	\$92.64	0.14	\$12.97	COD	\$159.34	\$482.75	\$965.50	-\$806.16
4	\$139.05	0.29	\$40.32	COD	\$394.90	\$86.42	\$345.68	\$49.22
3	\$155.87	0.03	\$4.68	COD	\$453.58	\$386.56	\$1,159.68	-\$706.10
4	\$115.42	0.03	\$3.46	COD	\$447.83	\$419.28	\$1,677.12	-\$1,229.29
3	\$593.62	0.08	\$47.49	Credit Card	\$1,638.39	\$212.26	\$636.78	\$1,001.61
2	\$825.04	0.20	\$165.01	Credit Card	\$1,320.06	\$318.91	\$637.82	\$682.24
2	\$471.88	0.24	\$113.25	PayPal	\$717.26	\$143.43	\$286.86	\$430.40
1	\$963.76	0.14	\$134.93	PayPal	\$828.83	\$174.02	\$174.02	\$654.81

Data Cleaning and Transformation

The following data cleaning activities were performed:

- Handled missing Cost values using Average replacement.
- Checked for missing and inconsistent data.
- Applied currency and percentage formatting.
- Integrated tables using XLOOKUP/VLOOKUP.
- Created calculated fields:
 - Discount Value
 - Total Cost
 - Total Profit
 - Age Group

=XLOOKUP(K2,Store_Dim!\$A:\$A,Store_Dim!\$C:\$C,"Not Found")

	F	G	H	I	J
Category	Sub Category	Age Group	Gender	Loyalty_Level	Region
	Team Sports	Adult	Male	Gold	North
Electronics	Audio	Senior Citizen	Female	Gold	North
	Outdoor	Young	Female	Silver	South
	Decor	Young	Female	Platinum	North

DATA CLEANING PROCESS

CONDITION APPLIED FOR AGE COLUMN

FORMULA=IF(C2>=60,"Senior Citizen",IF(C2>=30,"Adult","Young"))

Senior Citizen

Finding Missing Value in Loyalty Level Column

FORMULA=IF(ISBLANK(F2),"UNKNOWN",F2)

UNKNOWN

Total Cost	Total Profit
\$245.98	-\$47.56
\$1,211.13	-\$982.34
\$166.06	\$93.07
\$359.54	\$298.92
\$34.62	\$1,293.26
\$872.88	\$932.39
\$321.18	\$1,778.86
\$664.24	\$2,366.18
\$112.42	\$378.91
\$1,304.52	-\$615.35

Descriptive Statistics

Descriptive statistics were generated using Excel Analysis ToolPak, including:

- Mean
- Median
- Mode
- Standard Deviation
- Variance
- Kurtosis
- Skewness

<i>Discriptive Statistics for Total Profit</i>	
Mean	\$445.41
Standard Error	\$17.64
Median	\$349.38
Mode	#N/A
Standard Deviation	\$788.53
Sample Variance	Vertical (Value) Axis 1.09
Kurtosis	0.712730304
Skewness	0.590954694
Range	\$5,331.15
Minimum	-\$1,705.51
Maximum	\$3,625.65
Sum	\$890,368.85
Count	1999
Largest(1)	3625.648
Smallest(1)	-1705.5056
Confidence Level(95.0%)	34.58793745

Dashboard Components

The dashboard includes:

- Category-wise Sales and Profit Analysis
- Region-wise Sales and Profit Analysis
- Store Type Performance Analysis
- Monthly Sales Trend
- Interactive Slicers
- Key Performance Indicators(KPI)



Key Insights

- **Total Sales Revenue: \$2,170,907.46**
- **Total Profit: \$890,321.28**
- **South region generated the highest profit.**
- **Sports category generated the highest sales revenue.**
- **Customer purchasing behavior varies across loyalty levels.**
- **Sales performance differs among store types.**
- **Monthly sales trends reveal periods of higher and lower demand.**

SUMMARY REPORT				
1	Which category generated highest profit?	Sports = \$286,651.08		
2	Which category generated highest sales?	Sports = \$540,882.84		
3	Which region generated the highest profit ?	The South region generated the highest profit of \$273,292.68.		
4	Which region generated the lowest profit ?	The West region generated the lowest profit of \$112,007.24.		
5	What is the total profit earned by the business?	The total profit earned was \$890,321.28.		
6	What is the total sales revenue?	The total sales revenue was \$2,170,907.46.		
7	Which product category generated the highest sales?	The Sports category generated the highest sales revenue of approximately \$540,883.		
8	How many orders were processed?	A total of 2,000 orders were processed.		
9	Why was profit calculated?	Profit was calculated to measure the actual earnings from sales after deducting costs.		
10	Why was discount analyzed?	Discount analysis helps understand how price reductions influence sales and profitability.		

Business Questions Answered

Question	Answer
Which region generated the highest profit?	South Region
Which region generated the lowest profit?	West Region
Which category generated the highest sales?	Sports Category
What is the total sales revenue?	\$2,170,907.46
What is the total profit?	\$890,321.28
How many orders were processed?	2,000
How many customers were analyzed?	500

Formulas Used

CALCULATION	FORMULA
DISCOUNT VALUE	=Unit_Price*Discount
TOTAL COST	=Quantity*Cost
PROFIT	=Total_Amount-Total_Cost
TOTAL SALES	=SUM(Total_Amount)
AVERAGE SALES	=AVERAGE(Total_Amount)
TOTAL ORDERS	=COUNT(Sales_ID)
TOTAL CUSTOMERS	=COUNTA(Customer_ID)

Business Recommendations

- Increase investment in high-performing regions.
- Improve marketing activities in lower-performing regions.
- Maintain inventory levels for high-demand categories.
- Optimize discount strategies to improve profitability.
- Strengthen customer loyalty programs.
- Use dashboard insights for strategic planning and forecasting.

Conclusion

The project successfully demonstrated data cleaning, data integration, statistical analysis, Pivot Table reporting, and dashboard development using Microsoft Excel. The dashboard provides valuable business insights that support data-driven decision-making and improve organizational performance.

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Thank You